

# ENGINEERING MATERIALS

## **Q1. Explain Liquid Crystal Phases.**

In some of the substances, the tendency toward an ordered arrangement of molecules is so great that crystalline solid does not melt directly in to liquid state but passes first through an intermediate change that is called liquid crystal (or mesomorphic phase) before changing to liquid state on further application of heat.

The important property in liquid crystals is that small electric field can disturb the alignment of molecules while the large electric held induces turbulence in the liquid crystal with light scattering.

### **a. Smectic or soap like liquid crystals:**

A crystalline state where the orientation and periodicity characteristics are retained is called smectic liquid crystal. In other words, we can say the liquid crystals which have ordered position as well as orientation are smectic liquid crystals.



### **b. Nematic or thread like liquid crystals**

A crystalline state where the orientation is retained and periodicity characteristics are lost is called nematic liquid crystal. In other words, we can say the liquid crystals which have ordered orientation but disordered position is called nematic liquid crystals.



### **c. Cholesteric liquid crystals**

These liquid crystals have the same arrangement of molecules as that of Nematic type but their optical activity is many times higher than that of its solid crystalline variety. The cholesteric phase of liquid crystals has molecules parallel to each other but the direction of alignment twists gradually and results in a helical structure. Therefore, the substance consists of parallel layers. Molecules are aligned parallel in each layer. The helical structure of cholesteric substance is responsible for optical activity (optical rotation).



### **Application of Liquid Crystals**

- 1) Liquid crystals are used in Liquid Crystal Displays (LCD) like digital watches, pocket calculators because of the fact that their orientation can be easily changed by electric field or pressure.
- 2) When a beam of light strikes smectic liquid crystal planes, the properties of reflected beam depend upon the distance between planes. This distance between the planes is highly temperature sensitive and therefore color of the reflected light depends upon temperature. Arrangement based upon this principle can have sensitivity as good as  $0.01^{\circ}\text{C}$ .
- 3) Liquid crystals are used in gas-liquid chromatography because of electrical properties lying between crystalline solids and isotropic liquid.

### **Q2. Explain properties of Ferroics.**

1. Ferromagnetism: -A magnetism that is controlled by application of magnetic field.
2. Ferroelectricity: -An electric polarization that is switchable by applied electric field.
3. Ferroelasticity: -A deformation that is switchable by applied stress.

### **Q3. Explain Multiferroics and its type.**

Multiferroics are materials that exhibit more than one of the above primary properties of the Ferroic materials.

### Type I and Type II Multiferroics

Type I:→

These materials are the ones in which the ferroelectricity and ferromagnetism occur at different temperatures and arise from different mechanisms. Thus, ferroelectricity and ferromagnetism in such materials is largely independent of one another.

e.g.  $\text{YMnO}_3$ .

Type II:→

In these materials' magnetism causes ferroelectricity implying strong coupling between the two.

e.g.  $\text{TbMnO}_3$ .

### Applications of Multiferroics

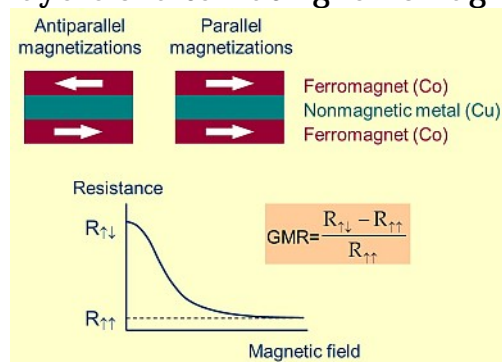
1. Electric field control of magnetism
2. Spintronic devices
3. Microelectronic devices MOSFET
4. Information storage devices
5. Quantum electromagnets in Particle physics
6. Sensors
7. Radio and high frequency devices

### Q4. What Magneto resistive materials.

The materials that have a tendency to change the value of their electrical resistance in an externally applied electric field.

### Q5. Explain GMR and CMR.

**GMR Giant Magnetoresistance:**→Magnetoresistance effect observed in layers of alternating ferromagnetic and non- magnetic conductive layers.



Electric resistance low for parallel alignment of adjacent ferromagnetic layers. Electric resistance high for antiparallel alignment of ferromagnetic layers.

### Applications

Sensors to read data, hard disc, drives, biosensors, MEMS (micro-electro-mechanical system), etc.

**CMR Colossal Magnetoresistance:**→Colossal MR is a property of some manganese-based perovskite oxides that enables them to drastically change their electrical resistance in presence of magnetic field which is strong.

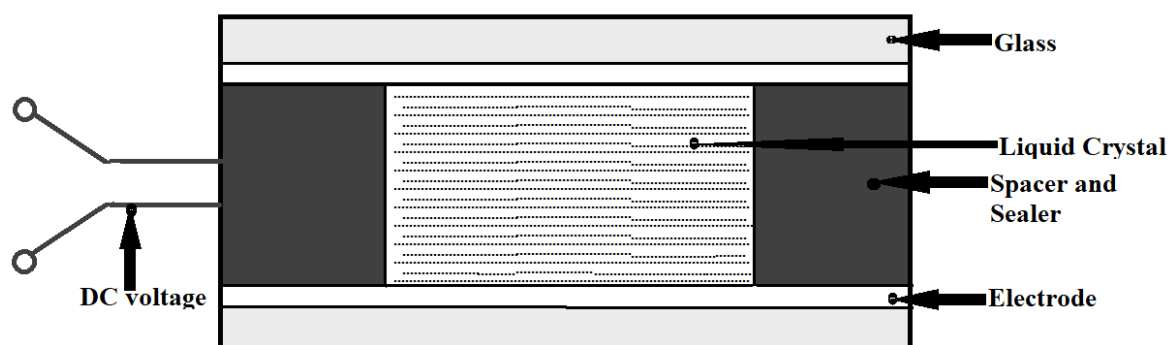
e.g.  $\text{LaMnO}_3$ .

### Applications

Read head devices, bolometers, spintronics, magnetic field sensors.

### Q6. Explain the working principle of Liquid crystal display.

Liquid crystal cell consists of a thin layer liquid crystal sandwiched between two glass sheets with transparent electrodes deposited on their inside face.



### Working:

The LCD does not produce any illumination of its own, but depends entirely on the illumination falling on it from an external source for its visual effect. We use cholesteric property of LCD. Thin Indium Tin oxide layer is coated over glass plate which acts as electrodes. If voltage is applied to the electrode, mesogens align themselves along field. The light waves get aligned along field. Hence, no transmission of light is possible.

**Merits:** LCD requires power in microwatts

**Demerit:** LCD is activated by AC and DC destroy display.

**Application:** Calculator, computer and TV.

### Q7. What is spintronics.

Spintronics is one of the most promising new generation information technology which uses the electron spin as the information carrier and

possess potential advantages of speeding up data processing and low energy consumption.

**Applications:**

Spin value

Datta Das spin transistor